

MEI YI YANG

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Education

University of Michigan

Ann Arbor, MI

Bachelor of Science in Engineering (BSE), Computer Science

August 2023 – May 2027

- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Distributed Systems, Web Systems, Machine Learning, Computer Organization, Statistics and Data Analytics, Linear Algebra, Game Engine Architecture

Experience

Frontend Research Assistant

Ann Arbor, MI

Radiological Health Engineering Lab, University of Michigan

May 2026 – Present

Technologies: React, Vite, Flask, Python, TypeScript, Next.js, MUI, Plotly.js, WordPress REST API

- Built a React/Vite dashboard visualizing 14+ live sensor channels (weather, radiation, enclosure) from Raspberry Pi hardware, porting existing Python rolling-mean and peak-hold smoothing algorithms into interactive Plotly.js charts
- Implemented an incremental live-polling pipeline with pagination and automatic outage detection, migrating the full data layer across two backend API redesigns with zero downtime
- Diagnosed and fixed a WebGL context-exhaustion crash across 12 concurrent charts by queuing renders off the main thread, and traced a live data-corruption bug to misrouted network traffic

AI Engineering Intern

Beijing, China

NetEase YouDao – R&D Team

May 2025 – August 2025

Technologies: PyTorch, Transformers/Hugging Face, Python, GPT-4/LLM APIs, CUDA, Pandas/NumPy

- Developed machine translation quality estimation framework for Tibetan-Chinese language pairs, implementing state-of-the-art metrics (COMET, chrF++, BERTScore, NLLB-200) to benchmark translation quality across 1,000+ test samples
- Built an automated LLM evaluation system using GPT-4 and Deepseek-v3, replacing most manual expert review through iterative prompt engineering
- Designed a statistical validation framework (Pearson, Spearman, Kendall's tau) benchmarking automated metrics against human judgments to identify translation model weaknesses

Operations Team Intern

Shenzhen, China

Tencent Cloud – CSIG Cloud Product Department

May 2024 – July 2024

Technologies: Python, Tencent HunYuan AI API, SQL

- Analyzed profitability data for international cloud products (CVM, Lighthouse, GPU) across 8 regions and proposed pricing optimizations for underperforming SKUs
- Conducted competitive analysis of AWS, Azure, and GCP pricing across 50+ instance types, identifying opportunities to offer equivalent performance at 5-10% lower cost while maintaining positive margins
- Deployed Tencent HunYuan AI chatbot for internal operations team, automating pricing calculations and reducing manual workload by 80%, saving 15 hours per week across 6 team members

Projects

Doodle FFA | *Typescript, SpacetimeDB, ElevenLabs, AWS* MLH Best Use of ElevenLabs – MHacks 2026

- Built a real-time multiplayer battle royale for 2-12 players with phones as controllers, syncing game state to every client from a 20 Hz server-authoritative simulation with reconnect handling
- Designed a generative AI pipeline where a vision LLM turns doodles into balanced weapons under a strict JSON schema, with server-side validation, ElevenLabs text-to-speech commentary, and fallbacks for every API call

Stillbite | *React Native, Expo, SQLite, RevenueCat, EAS*

- Built and shipped a full-stack iOS meal journal app from scratch to App Store, featuring a photo-first logging flow with a custom pinch-to-zoom crop modal, with raw image persistence in SQLite enabling non-destructive re-cropping
- Engineered a subscription layer with RevenueCat and StoreKit 2, gating PDF export, streak/fasting insights, and a regression-based cycle predictor behind a paywall enforced via React context

Thread Library | *C++, Multi-threading, Context Management, Unix*

- Implemented a user-level C++ thread library on Unix with multiprocessor support, interrupt handling, and FIFO scheduling; designed spin-locks, mutexes, and condition variables using Unix context management

Skills

Languages: C++, Python, Go, JavaScript, TypeScript, C#, C, Java, HTML/CSS, MySQL, SQLite

Frameworks & Libraries: ReactJS, React Native, Expo, Flask, PyTorch, Transformers, NumPy, Pandas, Jinja2, CI/CD, PixiJS

Tools & Platforms: Git, Linux/Bash, GitHub Actions, Docker, Unity, AWS, Vercel, Claude Code, agentic workflows, GDB, CUDA

Design Tools: Blender, Adobe Photoshop, GIMP, Aseprite